Guillarte, Dana Louise A.
S23A

Test Cases

Function	#	Description	Sample Input Data	Expected Output	Actual Output	P/F
promptChoice	1	Testing for a valid choice within the range	nMin = 1 nMax = 3 nChoice = 2	2	2	P
	2	Testing for a valid choice at the minimum of the range	nMin = 1 nMax = 3 nChoice = 1	1	1	P
	3	Testing for an invalid choice below the minimum of the range	nMin = 1 nMax = 3 nChoice = -1	Invalid input! Enter valid choice [1 - 3]: No return output	Invalid input! Enter valid choice [1 - 3]: No return output	P

login	1	Testing for a correct admin password on the first try	sAdminPass = “admin123”
			nRecords = 0
			records = “” //Array of empty question structures
			admin123 //Input via console
				Login successful!
				0
					Login successful!
					0
						P

	2	Testing for a correct admin password on the second try	sAdminPass = “admin123” nRecords = 5	Incorrect password. Press 1 to try again, or any other key	Incorrect password. Press 1 to try again, or any other key	P

			records = {...} //Array of 5 question structures hello //Input via console	to go back to main menu: 1 //Input via console Enter admin password: admin123 //Input via console Login successful! 5	to go back to main menu: 1 //Input via console Enter admin password: admin123 //Input via console Login successful! 5	
	3	Testing for an incorrect admin password	sAdminPass = “admin123” nRecords = 10 records = {...} //Array of 10 question structures good bye //Input via console	Incorrect password. Press 1 to try again, or any other key to go back to main menu: 0 //Input via console Going back to main menu	Incorrect password. Press 1 to try again, or any other key to go back to main menu: 0 //Input via console Going back to main menu	P
manageData	1	Testing adding a new record	nRecords = 0 records = “” //Empty array	nRecords = 1 records = {...} //Array with 1 new record added	nRecords = 1 records = {...} //Array with 1 new record added	P
	2	Testing deleting an existing record	nRecords = 2 records = {...} //Array with 2 records	nRecords = 1 records = {...} //Array with 1 record left after deletion	nRecords = 1 records = {...} //Array with 1 record left after deletion	P
	3	Testing importing data from	nRecords = 3 records =	nRecords > 3 records =	nRecords > 3 records =	P

		a file	{...} //Array with 3 records	{...} //Array with additional records imported from the file	{...} //Array with additional records imported from the file	
findRecord	1	Testing finding a record that exists at the beginning of the records array	records = [{sQText: "What is the smallest unit of life?", sChoices: ["Cell", "Atom", "Molecule"], nCorrAns: 1}, {sQText: "What is the process by which plants make their own food called?", sChoices: ["Photosynthesis", "Respiration", "Fermentation"], nCorrAns: 1}] nCount = 2 pQText = "What is the smallest unit of life?" pAnswer = "Cell"	nRecordInd = 0	nRecordInd = 0	P

	2	Testing finding a non-existing record	records = [{sQText: "What is the smallest unit of life?", sChoices: ["Cell", "Atom", "Molecule"], nCorrAns: 1}, {sQText: "What is the process by which plants make their own food called?", sChoices: ["Photosynthesis", "Respiration", "Fermentation"], nCorrAns: 1}] nCount = 2 pQText = "What is the smallest unit of life?" pAnswer = "Molecule"	nRecordInd = -1	nRecordInd = -1	P
	3	Testing finding a record that exists at somewhere between the beginning	records = [{sQText: "Which musical term refers to the gradual increase in	nRecordInd = 1	nRecordInd = 1	P

		and end of the records array	volume of a piece of music?", sChoices: ["Crescendo", "Decrescendo", "Staccato"], nCorrAns: 1}, {sQText: "Which composer is known for his 'Fifth Symphony' and 'Moonlight Sonata'?", sChoices: ["Ludwig van Beethoven", "Wolfgang Amadeus Mozart", "Johann Sebastian Bach"], nCorrAns: 1}] nCount = 2 pQText = "Which composer is known for his 'Fifth Symphony' and 'Moonlight Sonata'?" pAnswer = "Ludwig van			

			Beethoven"			
getLastQNum	1	Testing getLastQNum with a topic that has no records	records = [{sQText: "What is the capital of Canada?", sChoices: ["Toronto", "Montreal", "Ottawa"], nCorrAns: 3, sTopic: "Geography", nQNum: 1}, {sQText: "What is the smallest planet in our solar system?", sChoices: ["Earth", "Mars", "Mercury"], nCorrAns: 3, sTopic: "Science", nQNum: 2}] nCount = 2 pTopic = "Music"	nLastQNum = 0	nLastQNum = 0	P
	2	Testing getLastQNum with a topic that has one record	records = [{sQText: "Which musical term refers to the gradual increase in volume of a piece of music?",	nLastQNum = 1	nLastQNum = 1	P

			sChoices: ["Crescendo", "Decrescendo", "Staccato"], nCorrAns: 1, sTopic: "Music", nQNum: 1}] nCount = 1 pTopic = "Music"			
	3	Testing getLastQNum with a topic that has multiple records with different question numbers	records = [{sQText: "Which musical term refers to the gradual increase in volume of a piece of music?", sChoices: ["Crescendo", "Decrescendo", "Staccato"], nCorrAns: 1, sTopic: "Music", nQNum: 1}, {sQText: "Which composer is known for his "Fifth Symphony" and "Moonlight Sonata"?", sChoices:	nLastQNum = 3	nLastQNum = 3	P

			["Ludwig van Beethoven ", "Wolfgang Amadeus Mozart", "Johann Sebastian Bach"], nCorrAns: 1, sTopic: "Music", nQNum: 3}, {sQText: "What is the name of the force that opposes motion between two surfaces that are in contact?", sChoices: ["Friction", "Gravity", "Magnetism"], nCorrAns: 1, sTopic: "Science", nQNum: 2}] nCount = 3 pTopic = "Music"			
addRecord	1	Testing addRecord with a new record to a topic that has no existing records	records = [] sQText = "Who painted the Mona Lisa?" sChoices = ["Vincent van Gogh",	records = [{sQText: "Who painted the Mona Lisa?", sChoices: ["Vincent van Gogh",	records = [{sQText: "Who painted the Mona Lisa?", sChoices: ["Vincent van Gogh",	P

			"Pablo Picasso", "Leonardo da Vinci"] nCorrAns = 3 sTopic = "Art" nCount = 0	"Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 3, sTopic: "Art", nQNum: 1}] nCount = 1	"Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 3, sTopic: "Art", nQNum: 1}] nCount = 1	
	2	Testing addRecord with a new record to a topic that has existing records	records = [{sQText: "Who painted the Mona Lisa?", sChoices: ["Vincent van Gogh", "Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 3, sTopic: "Art", nQNum: 1}, {sQText: "Who painted the Starry Night?", sChoices: ["Vincent van Gogh", "Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 1, sTopic: "Art", nQNum: 2}] sQText = "Who painted the Sistine	records = [{sQText: "Who painted the Mona Lisa?", sChoices: ["Vincent van Gogh", "Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 3, sTopic: "Art", nQNum: 1}, {sQText: "Who painted the Starry Night?", sChoices: ["Vincent van Gogh", "Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 1, sTopic: "Art", nQNum: 2}, {sQText: "Who painted the Sistine	records = [{sQText: "Who painted the Mona Lisa?", sChoices: ["Vincent van Gogh", "Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 3, sTopic: "Art", nQNum: 1}, {sQText: "Who painted the Starry Night?", sChoices: ["Vincent van Gogh", "Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 1, sTopic: "Art", nQNum: 2}, {sQText: "Who painted the Sistine	P

			Chapel ceiling?" sChoices = ["Raphael", "Michelangelo", "Leonardo da Vinci"] nCorrAns = 2 sTopic = "Art" nCount = 2	Chapel ceiling?", sChoices: ["Raphael", "Michelangelo", "Leonardo da Vinci"], nCorrAns: 2, sTopic: "Art", nQNum: 3}] nCount = 3	Chapel ceiling?", sChoices: ["Raphael", "Michelangelo", "Leonardo da Vinci"], nCorrAns: 2, sTopic: "Art", nQNum: 3}] nCount = 3	
	3	Testing addRecord with a new record duplicate of an existing record	records = [{sQText: "Who painted the Mona Lisa?", sChoices: ["Vincent van Gogh", "Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 3, sTopic: "Art", nQNum: 1}, {sQText: "Who painted the Starry Night?", sChoices: ["Vincent van Gogh", "Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 1, sTopic: "Art", nQNum: 2}]	Record Already Listed: ---------- Topic: Art Question: Who painted the Starry Night? Choices: 1. Vincent Van Gogh 2. Pablo Picasso 3. Leonardo da Vinci Correct Answer: 1	Record Already Listed: ---------- Topic: Art Question: Who painted the Starry Night? Choices: 4. Vincent Van Gogh 5. Pablo Picasso 6. Leonardo da Vinci Correct Answer: 1	P

			sQText = "Who painted the Starry Night?" sChoices = ["Vincent van Gogh", "Pablo Picasso", "Leonardo da Vinci"] nCorrAns = 1 sTopic = "Art" nCount = 2			
removeNewLine	1	Testing removeNewLine with a string that has no newline characters	pStr = "Vincent Van Gogh"	pStr = "Vincent Van Gogh"	pStr = "Vincent Van Gogh"	P
	2	Testing removeNewLine with a string that has a single newline character at the end	pStr = "Pablo Picasso\n"	pStr = "Pablo Picasso"	pStr = "Pablo Picasso"	P
	3	Testing removeNewLine with a string that has multiple newline characters	pStr = "Leonardo da Vinci\n\n"	pStr = "Leonardo da Vinci"	pStr = "Leonardo da Vinci"	P
validateAnswer	1	Testing validateAnswer with a valid answer for	record = {sQText: "What is the capital of	nFoundIndex = 2	nFoundIndex = 2	P

		a multiple choice question	Canada?", sChoices: ["Toronto", "Montreal", "Ottawa"], nCorrAns: 2, sTopic: "Geography", nQNum: 1} sNewAns = "Ottawa"			
	2	Testing validateAnswer with an invalid answer for a multiple choice question	record = {sQText: "What is the capital of Canada?", sChoices: ["Toronto", "Montreal", "Ottawa"], nCorrAns: 2, sTopic: "Geography", nQNum: 1} sNewAns = "Calgary"	nFoundIndex = -1	nFoundIndex = -1	P
	3	Testing validateAnswer with an answer that is the correct answer for a multiple choice question	records = {sQText: "Who painted the Mona Lisa?", sChoices: ["Vincent van Gogh", "Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 3, sTopic: "Art", nQNum: 1} sNewAns =	nFoundIndex = 2	nFoundIndex = 2	P

			“Leonardo da Vinci”			
editRecord	1	Testing ability to edit a choice of a record	records = {sQText: "Who painted the Mona Lisa?", sChoices: ["Vincent van Gogh", "Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 3, sTopic: "Art", nQNum: 1} nQCount = 1 Enter choice index to edit (1 - 3): 1 //Input via console Enter new choice 1: Hello //Input via console	records = {sQText: "Who painted the Mona Lisa?", sChoices: ["Hello", "Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 3, sTopic: "Art", nQNum: 1}	records = {sQText: "Who painted the Mona Lisa?", sChoices: ["Hello", "Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 3, sTopic: "Art", nQNum: 1}	P
	2	Testing ability to edit an answer of a record	records = {sQText: "Who painted the Mona Lisa?", sChoices: ["Vincent van Gogh", "Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 3, sTopic: "Art",	records = {sQText: "Who painted the Mona Lisa?", sChoices: ["Vincent van Gogh", "Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 1, sTopic: "Art",	records = {sQText: "Who painted the Mona Lisa?", sChoices: ["Hello", "Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 1, sTopic: "Art", nQNum: 1}	P

			nQNum: 1} nQCount = 1 Enter new answer: “Vincent van Gogh” //Input via console	nQNum: 1}		
	3	Testing ability to edit a topic of a record	records = {sQText: "Who painted the Mona Lisa?", sChoices: ["Vincent van Gogh", "Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 3, sTopic: "Art", nQNum: 1} nQCount = 1 Enter new topic: Test //Input via console Enter new choice 1: Hello //Input via console	records = {sQText: "Who painted the Mona Lisa?", sChoices: ["Hello", "Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 3, sTopic: "Test", nQNum: 1}	records = {sQText: "Who painted the Mona Lisa?", sChoices: ["Hello", "Pablo Picasso", "Leonardo da Vinci"], nCorrAns: 3, sTopic: "Test", nQNum: 1}	P
deleteRecord	1	Testing deleteRecord with a valid input where the user confirms the deletion	records = [{sQText: "What is the capital of France?", sChoices: ["Berlin", "Paris", "Rome"], nCorrAns:	3	3	P

			2, sTopic: "Geography", nQNum: 1}, {sQText: "What is the highest mountain in the world?", sChoices: ["K2", "Mount Everest", "Makalu"], nCorrAns: 2, sTopic: "Geography", nQNum: 2}, {sQText: "What is the largest country by land area?", sChoices: ["Russia", "China", "United States"], nCorrAns: 1, sTopic: "Geography", nQNum: 3}, {sQText: "What is the currency of Japan?", sChoices: ["Yen", "Dollar", "Euro"], nCorrAns: 1, sTopic:			

			"Geography", nQNum: 4}] nCount = 4 Topic index: 1 Question index: 1 Confirmation: 1			
	2	Testing deleteRecord with a valid input where the user cancels the deletion	records = [{sQText: "What is the capital of France?", sChoices: ["Berlin", "Paris", "Rome"], nCorrAns: 2, sTopic: "Geography", nQNum: 1}, {sQText: "What is the highest mountain in the world?", sChoices: ["K2", "Mount Everest", "Makalu"], nCorrAns: 2, sTopic: "Geography", nQNum: 2}, {sQText: "What is the largest country by land	4	4	P

			area?", sChoices: ["Russia", "China", "United States"], nCorrAns: 1, sTopic: "Geography", nQNum: 3}, {sQText: "What is the currency of Japan?", sChoices: ["Yen", "Dollar", "Euro"], nCorrAns: 1, sTopic: "Geography", nQNum: 4}] nCount = 4 Topic index: 1 Question index: 1 Confirmation: 2			
	3	Testing deleteRecord with an invalid topic index input	records = [{sQText: "What is the capital of France?", sChoices: ["Berlin", "Paris", "Rome"], nCorrAns: 2, sTopic: "Geography", nQNum:	Invalid topic index Do you want to delete another record? (1-Yes, 2-No): 2 nCount = 4	Invalid topic index Do you want to delete another record? (1-Yes, 2-No): 2 nCount = 4	P

			1}, {sQText: "What is the highest mountain in the world?", sChoices: ["K2", "Mount Everest", "Makalu"], nCorrAns: 2, sTopic: "Geography", nQNum: 2}, {sQText: "What is the largest country by land area?", sChoices: ["Russia", "China", "United States"], nCorrAns: 1, sTopic: "Geography", nQNum: 3}, {sQText: "What is the currency of Japan?", sChoices: ["Yen", "Dollar", "Euro"], nCorrAns: 1, sTopic: "Geography", nQNum: 4}]			

			nCount = 4 Invalid Input: nTopicInd = 5 nQstInd = 2 nConfirm = 1 nChoice = 2			
readData	1	Testing with a valid file name	sFileName = "GuillarteDImport" records = an empty array of structs of question type nMaxQst = MAX_RECORDS nRecords = 0	nRecords > 0	nRecords > 0	P
	2	Testing with an invalid file name	sFileName = "import" records = an empty array of structs of question type nMaxQst = MAX_RECORDS nRecords = 0	nRecords = -1	nRecords = -1	P
	3	Testing with a file name with file extension	sFileName = "GuillarteDImport.txt" records = an empty array of structs of	nRecords = -1	nRecords = -1	P

			question type nMaxQst = MAX_RECORDS nRecords = 0			
writeData	1	Testing with a valid file name	sFileName = "export" records = an array of structs of question type, containing 2 question records with different topics, question numbers, question text, answer choices, and correct answers nQCount = 2	Exported sorted export.txt	Exported sorted export.txt	P
	2	Testing with an file name with file extension	sFileName = "export.txt" records = an array of structs of question type, containing 2 question records with different topics, question numbers, question text,	Exported sorted export.txt.txt	Exported sorted export.txt.txt	P

			answer choices, and correct answers nQCount = 2			
	3	Testing when file fails to open	sFileName = "export" records = an array of structs of question type, containing 2 question records with different topics, question numbers, question text, answer choices, and correct answers nQCount = 3	Failed to open file.	Failed to open file.	P
importData	1	Testing with a valid file name	sFileName = "GuillarteDImport" records = an empty array of structs of question type nMaxQst = MAX_RECORDS nRecords = 0	nRecords > 0	nRecords > 0	P
	2	Testing with an invalid file	sFileName = "import" records =	nRecords = -1	nRecords = -1	P

		name	an empty array of structs of question type nMaxQst = MAX_RECORDS nRecords = 0			
	3	Testing with a file name with file extension	sFileName = "GuillarteDImport.txt" records = an empty array of structs of question type nMaxQst = MAX_RECORDS nRecords = 0	nRecords = -1	nRecords = -1	P
exportData	1	Testing with a valid file name	sFileName = "export" records = an array of structs of question type, containing 2 question records with different topics, question numbers, question text, answer choices, and correct answers nQCount = 2	Successfully exported data to export.txt.	Successfully exported data to export.txt.	P

	2	Testing with an file name with file extension	sFileName = "export.txt" records = an array of structs of question type, containing 2 question records with different topics, question numbers, question text, answer choices, and correct answers nQCount = 2	Successfully exported data to export.txt.txt.	Successfully exported data to export.txt.txt.	P
	3	Testing when file fails to open	sFileName = "export" records = an array of structs of question type, containing 2 question records with different topics, question numbers, question text, answer choices, and correct answers nQCount = 3	Failed to export data.	Failed to export data.	P
playGame	1	Testing	records =	John’s new	John’s new	P

		with valid input where the user selects a topic and answers a question correctly	array with 3 records nQCount = 3 scores = empty array *pScoreCount = 0	score: 1	score: 1	
	2	Testing with valid input where the user has played before, selects a topic, and answers a question correctly	records = array with 3 records nQCount = 3 scores = empty array *pScoreCount = 3	John's new score: 4	John's new score: 4	P
	3	Testing with valid input where the user selects a topic and answers a question incorrectly	records = array with 3 records nQCount = 3 scores = empty array *pScoreCount = 0	John's new score: 0	John's new score: 0	P
loadScores	1	Testing with no score file yet	scores = empty array	No scores found.	No scores found.	P
	2	Testing with score file with 1 player	scores = array with 1 player score	nReadInd = 1	nReadInd = 1	P
	3	Testing with score file with more than 1 player	scores = array with 2 player scores	nReadInd = 2	nReadInd = 2	P

saveScores	1	Testing with no score file cannot be found	scores = array with 1 player score nTotalScores = 1	Error writing scores.	Error writing scores.	P
	2	Testing with score file with 1 player	scores = array with 1 player score nTotalScores = 1	John 1	John 1	P
	3	Testing with score file with more than 1 player	scores = array with 2 player scores nTotalScores = 2	John 2 Dana 1	John 2 Dana 1	P